The Course TA-Bot

ECE 6256 / CS 6604 · Assignment 3 · Conversational retrieval with filtered user memory

1. System design

The bot searches 108 course papers and 5,711 supplied chunks through SQLite FTS5. Quoted, stopword-filtered query terms form a lexical OR search. BM25 gives title terms weight 5; the top 40 chunks are grouped into papers, with up to two chunks per paper. Paper scores sum the selected chunk scores. Sorting uses explicit identifiers to break ties.

For follow-ups, a course-endpoint completion rewrites references into a standalone query. Reciprocal rank fusion combines original and rewritten results (damping 60), retaining the original ranking on ties. Pronouns retain the last subject; explicit comparisons can add earlier primary papers. Topic switches skip rewriting. The last three exchanges provide context. Generation receives attributed passages, instructions to cite paper IDs, and a separate description of stored user facts for personalization. All inference goes through the configured endpoint; no external model API or local inference is used.

2. Controlled ablation: papers retrieved per query

Only retrieval depth k changes (1, 2, 3, 5, 8). The corpus, chunking, BM25 weights, prompts, rewriting, seed 42, and temperature 0 remain fixed. Each setting runs twice with fresh bot state on the 10 public single-turn questions and 3 dialogues (10 turns). Gold paper IDs are consulted after generation only. Dialogue-end recall matches eval_dev.py; all-turn recall is an additional diagnostic. Evidence size includes passage headers and measures prompt cost in characters, rather than model tokens.

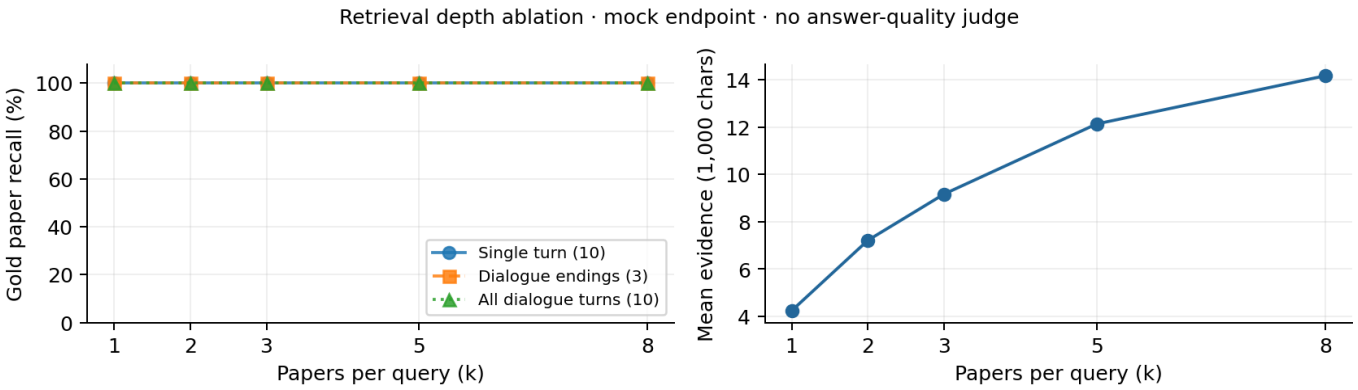


Figure 1. Bundled mock endpoint. Curves overlap at 100% recall; larger k increases evidence size. Prior subjects can make returned paper counts exceed k. The mock generates canned text and provides no answer-quality measurement.

k	Single recall	Dialogue final	All dialogue turns	Mean evidence chars
1	100%	100%	100%	4,242
2	100%	100%	100%	7,219
3	100%	100%	100%	9,177
5	100%	100%	100%	12,139
8	100%	100%	100%	14,178

3. Interpretation and limits of the ablation

The public retrieval task is already saturated at k=1: every tested setting achieves 100% single-turn recall, dialogue-ending recall, and all-turn dialogue recall. Mean evidence grows from 4,242 characters at k=1 to 14,178 at k=8 (3.34 times). At the existing default k=3, the mean is 9,177 characters. These observations show additional prompt cost without a measured public-set recall benefit. They do not establish which k produces better explanations.

The existing default k=3 is retained as an unvalidated recall cushion, not an empirically optimal choice. A real-endpoint comparison should judge answer accuracy and grounding at k=1 versus k=3 before choosing between them. The sample has only 10 independent single-turn items and 3 independent dialogues; dialogue turns are correlated. Broad lexical matching plus context carry-over can supply irrelevant papers while still satisfying recall. Returned citations list evidence supplied to generation; their presence alone does not prove that generated claims used or correctly interpreted it.

4. Memory design and privacy reasoning

What is stored. A per-user SQLite table stores readable category/fact pairs. An allowlist accepts explicit first-person statements about a name, project or thesis, academic background, interests, and response preferences. Storage occurs on user turns before retrieval, never from assistant text or paper passages. Duplicate facts are ignored. New names and projects replace older values; interests, preferences, and backgrounds can coexist. At most 30 facts per user are retained, evicting older rows when needed. Without memory_path, the SQLite store lives in process memory. Setting memory_path retains facts across process restarts; session history remains process-local.

What is refused. A whole-message opt-out overrides extraction. Per-sentence vetoes reject secrets and identifiers, contact details, health and protected attributes, financial and academic-standing information, third-party statements, hypothetical or negated claims, temporary states, questions, and recognizable instruction injection. Anchoring patterns to first-person clauses prevents quoted third-party examples from becoming user identity. Filtering a whole sensitive sentence favors privacy over remembering a safe preference mixed into that sentence. Explicit requests to remember something do not override these refusals.

Isolation and deletion. Facts are keyed by user_id, while history uses both user_id and session_id. Fresh sessions share only that user's accepted facts. Direct memory questions read the store without generation, preventing invented recall. Explicit forgetting deletes the requested category or all facts and clears that user's histories, so deleted information is not replayed in later prompts. reset_memory performs the same full deletion. Refused messages and their responses are redacted from replayed history.

Tradeoffs. English regular expressions are conservative, not a semantic privacy guarantee. They can miss paraphrases and reject harmless academic language. Preferences can conflict because only names and projects have replacement semantics. A persistence file is ordinary SQLite, without encryption or authentication; callers must supply a trusted user identity and protect the file. Sensitive text is still sent to the configured course endpoint for its current response; refusal means no retained profile fact or replayed conversation text, not that the endpoint never sees the message.

5. Validation and reproducibility

Check	Observed result
Public single-turn retrieval	10 / 10 gold papers found
Public dialogue endings	3 / 3 dialogues at full recall
Public positive memory probes	2 / 2; all listed expected facts checked
Additional authored probes	2 / 2 positive; 8 / 8 negative
Regression suite	35 / 35 tests pass
Ablation repeated outputs	Identical answers, citations, evidence at every k

Regressions cover valid paper-level citations, deterministic retrieval, hostile FTS syntax, endpoint URL normalization, malformed replies, timeout/error conversion, follow-up subjects, topic switches, user/session isolation, memory replacement, deduplication, persistence, opt-outs, sensitive-text replay, and full/category deletion. The bundled mock's hash-to-float conversion was repaired because arbitrary float bit patterns produced NaN embeddings; it now maps integer values to bounded floats.

The measured environment uses Python 3.8.5, SQLite 3.33.0, and requests 2.32.4. The fresh virtual environment installs requirements-dev.txt; requirements-lock.txt records its exact package versions. TABot's runtime dependencies are listed separately in requirements.txt. The saved results include per-turn predictions, answers, evidence lengths, timings, environment versions, source/corpus/dataset SHA-256 checksums, and both repeated runs. No endpoint address or credential is embedded in the implementation.

```
python -m pip install -r requirements-dev.txt
python scripts/verify.py
python experiments/ablation.py --mock
python scripts/build_report.py
python scripts/build_submission.py
```

To test the real course model, run eval_dev.py with --endpoint and --model, or run experiments/ablation.py with the same arguments and --output results-course. Keep real-model results separate from this mock study. Endpoint failures become TimeoutError rather than unhandled transport or response-parsing exceptions.

6. Remaining evaluation limits

No real course vLLM endpoint was supplied during this run. Answer quality, hidden-set retrieval, real-model memory leakage, graded-server latency, and the stated 0.76 quality threshold are unmeasured. The added negative probes are authored development checks, not hidden tests. Repeated mock outputs establish local determinism, not determinism of a remote model server. No LLM judge was run, so no judge-error bonus case is claimed.

7. AI-use disclosure

OpenAI Codex was used to inspect and extend the supplied implementation, diagnose compatibility and privacy problems, write code and regression probes, run the local evaluation and retrieval-depth ablation, generate the plot and this report, and prepare the submission archive. The supplied repository already contained retrieval, generation, rewriting, and memory code; earlier tool usage was unavailable in this session. This disclosure covers the continuation. Verification was automated through a fresh Python environment, the bundled mock, public dev data, authored tests, and repeated ablation runs. Measurements came from executed code, with raw records retained. No human review, real-model answer evaluation, or hidden-set result is represented as completed.